Information about Physical Activity Guidelines, Sleep Guidelines, Cardiorespiratory Fitness, and Additional Resources

Health, Brain & Cognition Lab

Physical Activity Guidelines

The Office of Disease Prevention and Health Promotion recommends 150 to 300 minutes of moderate physical activity per week OR 75 to 150 minutes of vigorous activity per week OR any combination of the two. Adults who sit less and do any amount of moderate to vigorous physical activity gain some health benefits. Link: https://tinyurl.com/odphpguidelines The Canadian 24-hour Movement Guidelines recommend limiting the amount of time spent sitting, or sedentary, to less than 8 hours a day and replacing sedentary time with light activity or bouts of standing. The guidelines also recommend several hours of light physical activity per day. Link: https://tinyurl.com/ CSEPpaguidelines

Sleep Guidelines

The CDC recommends that adults sleep at least 7 hours each night. Sleep efficiency, an indicator of good sleep quality, is recommended to be at least 85% by the National Sleep Foundation. This can be calculated by dividing the time spent asleep by the time spent in bed, then multiplied by 100. Link: https://www.cdc.gov/sleep/about/index.html

Cardiorespiratory Fitness

Cardiorespiratory fitness (CRF) is the volume of oxygen (mL) per body weight (kg) that you consume per minute when working at your maximum level. Our exercise test estimates VO_2max, which is a measure of CRF. You can think of CRF as your "aerobic engine size". A CRF greater than 20 ml/kg/min is associated with 1-2x lower dementia risk in the next 10-20 years while a CRF of around 30 ml/kg/min is associated with good brain outcomes in age groups of 50+ years. Since our exercise test estimates CRF, there is a certain margin of error associated with the value. Sources of error can include variability in age-predicted heart rate max, assuming a linear relationship exists between heart rate and work rate, variability in heart rate at the same work rate on different days, certain medications, and caffeine consumption.

Additional Resources

If you want to estimate your CRF on your own, try this non-exercise equation https://michelle-voss.shinyapps.io/resources_estimated-crf/ or this simple exercise-based field test https://tinyurl.com/rockportwalkingtest. Check out https://chtc.sites.uiowa. edu/health-coaching for health coaching services and https://chtc.sites.uiowa. edu/health-testing-services for health testing services through the University of Iowa.